

Graph Caption Framing Classroom Run Kit

TOK Cognitive Science Activity Bank • Knowledge and Language • 20-30 min

Category	Knowledge and Language	Time	20-30 min
Difficulty	Easy	ID	graph-caption-framing

Big idea: Data needs language before it becomes a public knowledge claim.

One-Minute Teacher Brief

Students write different captions for the same graph and analyze how language changes what the data seems to show.

Student Prompt

Inspect Graph Caption Framing Stimulus A. Record one first claim, the evidence you used, your confidence, and what would make you revise.

Concrete Stimulus Packet

Graph Caption Framing Stimulus A	Student-facing stimulus 1: Line graph: screen time rises during exam week and falls afterward. Annotate the wording. Circle the words that change tone, emotion, implication, or trust.	Students inspect this exact card first and commit to a claim before hearing anyone else.
Graph Caption Framing Stimulus B	Student-facing stimulus 2: Caption contrast: 'Students lose balance during exams' versus 'Temporary study routines shift during assessment week'. Annotate the wording. Circle the words that change tone, emotion, implication, or trust.	Use this for comparison, a second round, or a partner challenge.
Graph Caption Framing Reveal / Twist	Student-facing stimulus 3: Caption contrast: 'Students lose balance during exams' versus 'Temporary study routines shift during assessment week'. Annotate the wording. Circle the words that change tone, emotion, implication, or trust.	Teacher reveal: connect the changed response to the big idea — Data needs language before it becomes a public knowledge claim.

Actual Lesson Questions

#	Question for students	Teacher key / cue
1	After inspecting Graph Caption Framing Stimulus A, what is your first knowledge claim?	Teacher cue: look for a clear claim, not just a description.
2	Which exact detail from Graph Caption Framing Stimulus A did you use as evidence?	Teacher cue: students should name visible words, data, source features, method features, or contextual clues.
3	What did you infer beyond the evidence?	Teacher cue: this is where assumptions, framing, models, or perspective usually appear.
4	How confident are you: 50%, 60%, 70%, 80%, 90%, or 100%?	Teacher cue: confidence is data for the debrief, not a grade.
5	What missing information would most improve the knowledge claim?	Teacher cue: push for method, source, context, comparison, or definitions.
6	When does language become part of evidence?	Teacher cue: students should move from the classroom example to a general TOK issue.

Student Task

Use the stimulus cards to complete the observation/inference/evidence/confidence grid, then answer one TOK knowledge question connected to Knowledge and Language.

Teacher Key / Reveal

The key move is not the “right answer”; it is the moment students see how data needs language before it becomes a public knowledge claim.

- Strong response: separates observation from inference.
- Strong response: names a method, source, model, language choice, context, or perspective.
- Strong response: revises confidence when new evidence or a new criterion appears.
- Weak response: gives only opinion or says “it depends” without criteria.

Student Instructions

- Work silently for the first response so you can notice your own starting point.
- Record one knowledge claim, the evidence behind it, and your confidence level.
- After the reveal or comparison, update your claim in a different colour or column.
- Use the debrief to answer: When does language become part of evidence?

Setup Checklist

- Use a graph simple enough that students can understand it quickly.
- Remove the original title and caption so students experience the framing gap.

Example Stimuli or Materials

- Line graph: screen time rises during exam week and falls afterward.
- Caption contrast: 'Students lose balance during exams' versus 'Temporary study routines shift during assessment week'.

Resources To Use

- Resource pack: <resources/graph-caption-framing/index.html>
- Card deck CSV: <resources/graph-caption-framing/cards.csv>
- Visual prompt: <resources/graph-caption-framing/visual-prompt.svg>
- Caption frame cards: cautious, alarming, celebratory, skeptical, policy-focused.
- Graph context checklist: sample, scale, missing variables, time period, comparison group.

Run Sequence

1. Show the graph without title or caption and ask students what it might show.
2. Groups write captions in assigned framing styles.
3. Students compare how each caption changes interpretation and trust.
4. Groups identify what extra context the graph needs.
5. Debrief the difference between data, interpretation, and persuasion.

Debrief Moves

- Knowledge question: When does a caption become part of the evidence?
- Can accurate language still mislead?

- What responsibility do communicators have when turning data into claims?

Teacher Quick Script

- The graph is not silent; its labels teach us how to read it.
- What does your caption invite the reader to conclude?

Exit Ticket

- One thing I first treated as evidence was...
- My confidence changed because...
- A better knowledge question I can now ask is...